

Swahili Proverb Classification

Application of Natural Language Processing Classification Methods for Low-Resource and Agglutinative Languages

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Abstract

This paper evaluates natural language processing methods for classifying 5,060 Swahili proverbs into 56 semantic categories. Swahili remains significantly under-resourced in NLP despite an estimated 200 million speakers across East and Central Africa. The task is complicated by metaphorical language, severe class imbalance, and Swahili's agglutinative morphology. Four paradigms are compared: static embeddings (TF-IDF, character n-gram TF-IDF, Word2Vec, FastText) fed into a shallow neural network; fine-tuning of DistilBERT-multilingual, SwahBERT, and AfriBERTa-Large; masked language model domain adaptation prior to fine-tuning; and zero-shot and few-shot prompting of Llama-3-8B-Instruct. Results are reported using Macro-F1 and MCC given the class imbalance. Contrary to expectations, sparse statistical representations outperformed all neural approaches: character n-gram TF-IDF achieved the highest scores (accuracy 0.231, Macro-F1 0.225, MCC 0.211). AfriBERTa-Large was the strongest transformer (Macro-F1 0.116), MLM adaptation produced consistent gains for both models, and Llama-3 zero-shot was surprisingly competitive on MCC (0.212). These results suggest that surface and sub-word patterns generalise better than out-of-domain contextual representations on small, lexically sparse corpora.

Keywords: *Swahili, proverb classification, TF-IDF, character n-grams, SwahBERT, AfriBERTa, low-resource NLP.*

1. Introduction

"Haba na haba hujaza kibaba" is a Swahili proverb meaning that small and consistent efforts accumulate into meaningful outcomes. This proverb reflects a working principle in machine learning: models improve progressively and incrementally through exposure to data and better representations. Natural language processing (NLP), as a branch of machine learning concerned with how computers represent and interpret human language, embodies this same principle. Progress on a task like Swahili proverb classification depends significantly on how well the underlying text is represented, and these representations improve incrementally as methods advance from simple word counts to richer contextual models.

Swahili is one of Africa's most widely spoken languages, serving as a lingua franca across East and Central Africa with an estimated 200 million speakers (Ethnologue, 2023). Despite this wide reach, Swahili remains significantly under-represented in NLP research, with fewer dedicated resources and benchmark datasets compared to high-resource languages such as English, French, or Mandarin. Proverbs constitute a particularly rich form of Swahili language use: they encode accumulated wisdom through concise, metaphorical expressions that bear no surface-level semantic resemblance to their underlying meaning. A proverb categorised under 'Debt/Borrowing' does not typically contain the Swahili word for borrowing; instead, it may describe a falling tree, a setting sun, or a broken bag. This semantic gap makes proverb classification a demanding test of a model's capacity for deep semantic understanding.

The challenge is compounded by Swahili's agglutinative morphology. Unlike English, where grammatical information is largely conveyed through separate function words, Swahili encodes tense, subject, object, and mood through affixes attached to a single verb stem, so a single Swahili token can carry the semantic load of an entire English phrase. This paper is a comparative study of four NLP methods: classical static embedding methods (TF-IDF, character n-grams, Word2Vec, FastText), transformer fine-tuning (DistilBERT-multilingual, SwahBERT, AfriBERTa-Large), MLM domain adaptation, and large language model prompting (Llama-3-8B-Instruct). The study addresses four research questions:

1. Which performs best on this low-resource, metaphorically rich classification task?
2. Does language-specific pre-training achieve a measurable advantage over multilingual models?
3. Does MLM domain adaptation on the target corpus provide further improvement?
4. Can a large language model perform competitively without task-specific training, under zero-shot and few-shot conditions?

2. Related Works

Text classification has evolved from sparse statistical representations (Sparck Jones, 1972) through dense distributed embeddings (Mikolov et al., 2013; Bojanowski et al., 2017) to deep contextual representations introduced by BERT (Devlin et al., 2019) and its derivatives including DistilBERT (Sanh et al., 2019). For specialised domains, domain-adaptive pre-training (Gururangan et al., 2020), continued MLM pre-training on the target domain before fine-tuning, has been established as a reliable low-resource technique. Large

language models such as Llama-3 (Dubey et al., 2024) have introduced in-context learning as a new classification paradigm: providing task descriptions and optional examples in the prompt, without gradient updates. Few-shot prompting has achieved competitive results on many benchmarks (Brown et al., 2020), though performance on low-resource languages and culturally specific tasks remains less well characterised.

Research on NLP for African languages has expanded considerably in recent years. AfriBERTa (Ogueji et al., 2021) demonstrated that competitive multilingual language models can be trained on less than 1 GB of text across 11 African languages including Swahili, challenging the assumption that large pre-training corpora are a prerequisite for effective language models. AfroXLMR (Alabi et al., 2022) showed that multilingual adaptive fine-tuning of multilingual models on African language corpora substantially improves downstream task performance, suggesting that language and domain adaptation can partially compensate for a pre-training distribution dominated by high-resource languages. SwahBERT (Martin et al., 2022) provides a monolingual BERT for Swahili trained on news, online discussion, and Wikipedia text, motivating a direct comparison with multilingual alternatives: a dedicated monolingual model should better capture Swahili's morphological and syntactic patterns, though this advantage may diminish on a domain as semantically unusual as proverbs. To our knowledge, no prior study has systematically compared this full spectrum of NLP approaches on Swahili proverb classification.

3. Methodology

Dataset Description

The dataset consists of 5,060 Swahili proverbs organised into 56 semantic categories, sourced from the Swahili Proverbs collection of the Center for African Studies at the University of Illinois Urbana-Champaign (Scheven, 2010). Categories include abstract thematic labels such as Friendship, Debt/Borrowing, Work and Laziness, Poverty and Wealth, Speech and Silence, and Pride and Humility. The proverbs average approximately 5 to 9 words and 30 characters in length. The dataset shows significant class imbalance: some categories contain more than 200 examples while others have as few as 5. Data was split using stratified sampling with a fixed random seed (seed=42), ensuring reproducibility and balanced class representation, as shown in **Table 1**.

Split	Samples	Percentage
Training	3,643	~72%
Validation	405	~8%
Test	1,012	~20%
Total	5,060	100%

Table 1. Dataset split statistics.

Data Preprocessing

Minimal text processing was carried out given the average length of the proverbs. No stop words were removed; all text was lowercased and tokenised using whitespace and punctuation boundaries. No stemming or lemmatisation was applied: given Swahili's agglutinative morphology, aggressive stemming risks stripping meaningful grammatical affixes, such as tense prefixes *ali-* (past) and *ata-* (future), and noun class prefixes *m-*, *wa-*,

ki-, vi-, that may carry discriminative signal for certain semantic categories. For transformer models, tokenisation was handled by each model's own pre-trained tokeniser with a maximum sequence length of 64 tokens, which comfortably accommodates all proverbs in the corpus.

Evaluation Metrics

Given the significant class imbalance across the 56 categories, standard accuracy is a potentially misleading metric: a model predicting only majority classes can achieve high accuracy while failing on most underrepresented categories. Macro-F1 and Matthews Correlation Coefficient (MCC) are therefore adopted as primary metrics (Chicco and Jurman, 2020). Macro-F1 is the unweighted average of per-class F1 scores, treating all 56 categories equally regardless of support. MCC accounts for all four quadrants of the confusion matrix simultaneously and is directly interpretable: 0 represents random prediction, +1 perfect classification. Test accuracy is also reported throughout to enable consistent cross-task comparison.

Reproducibility and Tools

All experiments were implemented in Python and run in a Google Colab GPU environment. Classical baselines used scikit-learn for the TF-IDF and character n-gram vectorisers, train/validation/test splitting, and metric computation; Word2Vec embeddings were trained with gensim. Neural training used PyTorch, and all transformer and language models were loaded through the Hugging Face Transformers library. A fixed random seed of 42 was applied to Python, NumPy, and PyTorch throughout to support reproducibility. All training and evaluation runs were tracked with Weights & Biases (project: nalapro-swahili) and MLflow, logging per-epoch training loss, validation metrics, and final test accuracy, Macro-F1, and MCC for each experimental condition.

Task 1: Static Embeddings + MLP

Task 1 establishes baselines using static text representations combined with a shallow neural classifier. Five representation methods are evaluated, summarised in **Table 2**. TF-IDF (1-2gram) is a strong baseline for short-text classification because it down-weights common tokens and up-weights discriminative ones, although it is blind to morphological structure and cannot generalise to out-of-vocabulary words. Character n-gram TF-IDF applies TF-IDF at the character level (n-grams of length 3 to 6), which is well-motivated for Swahili because character n-grams implicitly capture morphological affixes. Word2Vec trains skip-gram embeddings of dimension 100 on the training corpus, with 1-epoch and 10-epoch variants; each proverb is represented as the mean of its token embeddings. FastText uses Facebook's pre-trained Swahili Common Crawl and Wikipedia vectors (cc.sw.300.bin), providing 300-dimensional embeddings with sub-word coverage (Grave et al., 2018). The classifier is a two-layer SimpleMLP with hidden dimension 128, ReLU activation, and cross-entropy loss, trained with Adam (lr=0.001, batch size=32) for 50 epochs. The checkpoint with highest validation Macro-F1 is retained for test evaluation.

Method	Type	Dimension	Pretrained on Swahili?
TF-IDF (1-2gram)	Sparse bag-of-words	1,000	No
Char N-gram TF-IDF	Sparse char n-grams (3 to 6)	2,000	No
Word2Vec (1 ep)	Dense word embeddings	100	No
Word2Vec (10 ep)	Dense word embeddings	100	No
FastText (cc.sw.300)	Dense subword embeddings	300	Yes

Table 2. Task 1 representation methods.

Task 2: Transformer Fine-Tuning

Task 2 fine-tunes transformer models for sequence classification, each with a classification head mapping the [CLS] token representation to 56 output logits, trained with AdamW, linear warmup, and gradient clipping at norm 1.0 (Devlin et al., 2019). The configurations are described in **Table 3**. DistilBERT-multilingual (distilbert-base-multilingual-cased) was fine-tuned for 10 epochs at $lr=3e-5$. SwahBERT (Martin et al., 2022), a monolingual Swahili BERT, was fine-tuned for 5 epochs at $lr=2e-5$; its vocabulary and tokeniser are natively designed for Swahili morphology, although proverbs are stylistically distant from the news-domain text that dominated its pre-training. AfriBERTa-Large (castorini/afriberta_large), a RoBERTa-style model pre-trained on 11 African languages including Swahili (Ogueji et al., 2021), was fine-tuned for 5 epochs at $lr=2e-5$.

Model	Architecture	Pretraining	Epochs	LR
DistilBERT-multilingual	6-layer distilled BERT	104 languages	10	$3e-5$
SwahBERT	12-layer BERT	Swahili (monolingual)	5	$2e-5$
AfriBERTa-Large	RoBERTa-style	11 African languages	5	$2e-5$

Table 3. Task 2 model configurations.

Task 3: MLM Domain Adaptation

Task 3 investigates whether continued MLM pre-training on the proverb corpus improves downstream classification, following the domain-adaptive pre-training paradigm (Gururangan et al., 2020). The full 5,060 proverb corpus is used for MLM adaptation (training, validation, and test combined), as labels are not used. This is a self-supervised procedure: 15% of tokens are dynamically masked per batch using Hugging Face's DataCollatorForLanguageModeling, and the model is trained to recover the original tokens. Adaptation runs for 1 epoch (AdamW, $lr=5e-5$, gradient clipping 1.0). AfriBERTa-Large and SwahBERT are each evaluated at baseline and post-adaptation, producing the four conditions shown in **Table 4**.

Condition	Base Model	MLM Adapted?	Cls Epochs
AfriBERTa-Baseline	castorini/afriberta_large	No	5
AfriBERTa-MLM	castorini/afriberta_large	Yes (1 ep, lr=5e-5)	5
SwahBERT-Baseline	SwahBERT	No	5
SwahBERT-MLM	SwahBERT	Yes (1 ep, lr=5e-5)	5

Table 4. Task 3 experimental conditions.

Task 4: LLM Prompting with Llama-3

Task 4 evaluates Llama-3-8B-Instruct (Dubey et al., 2024) under zero-shot and few-shot prompting on a random sample of 20 proverbs. For zero-shot classification, each proverb is presented with all 56 category names and a system prompt instructing the model to respond with only the category name, using the Llama-3 chat template. For few-shot classification, 10 randomly selected categories are each illustrated with one example proverb before the target is presented. Label extraction uses cascaded matching: exact string match, then substring match, then word-level overlap. Predictions matching no category are counted as invalid and excluded from metric computation. Because of the small sample, results at $n=20$ across 56 classes are necessarily noisy and should be treated as exploratory rather than definitive.

4. Results

Task 1: Static Embeddings

Results for all five static embedding methods are presented in **Table 5** and **Figure 1**. Character n-gram TF-IDF achieved the highest performance across every metric (test accuracy 0.231, Macro-F1 0.225, MCC 0.211), narrowly ahead of word-level TF-IDF (accuracy 0.213, Macro-F1 0.213) and pre-trained FastText (accuracy 0.227, Macro-F1 0.197). The strong showing of the character n-gram method supports the hypothesis that sub-word patterns are especially informative for an agglutinative language, since character n-grams implicitly capture morphological affixes. The two Word2Vec variants performed far worse (Macro-F1 0.009 and 0.005), effectively near random, indicating that embeddings trained from scratch on only 3,643 short proverbs do not acquire useful representations. Notably, the 10-epoch Word2Vec did not improve over the 1-epoch variant, consistent with overfitting on such a small corpus.

Method	Val Macro-F1	Test Accuracy	Test Macro-F1	Test MCC
TF-IDF (1-2gram)	0.197	0.213	0.213	0.193
Char N-gram TF-IDF	0.225	0.231	0.225	0.211
Word2Vec (1 ep)	0.014	0.057	0.009	0.062
Word2Vec (10 ep)	0.011	0.052	0.005	0.014
FastText (cc.sw.300)	0.210	0.227	0.197	0.206

Table 5. Task 1 results, static embeddings + MLP classifier. Best value per column in bold.

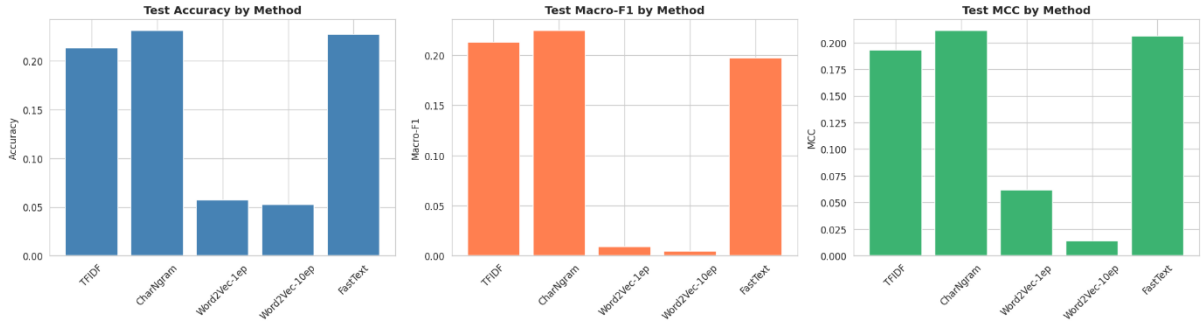


Figure 1. Test accuracy, Macro-F1, and MCC across Task 1 static embedding methods.

Task 2: Transformer Fine-Tuning

Results for the transformer fine-tuning experiments are shown in **Table 6** and **Figure 2**. AfriBERTa-Large achieved the best transformer performance on every metric (test accuracy 0.204, Macro-F1 0.116, MCC 0.180). DistilBERT-multilingual (Macro-F1 0.101) and SwahBERT (Macro-F1 0.070) trailed behind, and although SwahBERT and DistilBERT reached almost identical MCC (0.129), DistilBERT achieved a higher Macro-F1, indicating better coverage of low-support categories. Critically, the best transformer (AfriBERTa-Large, Macro-F1 0.116) underperformed the best static method from Task 1 (character n-gram TF-IDF, Macro-F1 0.225) by a wide margin, roughly half the Macro-F1, which is the central finding of this study.

Model	Val Macro-F1	Test Accuracy	Test Macro-F1	Test MCC
DistilBERT-multilingual (10ep)	0.086	0.155	0.101	0.129
SwahBERT	0.066	0.152	0.070	0.129
AfriBERTa-Large	0.111	0.204	0.116	0.180

Table 6. Task 2 results, transformer fine-tuning. Best value per column in bold.

Task 3: MLM Domain Adaptation

The effect of MLM domain adaptation is presented in **Table 7** and **Figure 3**. MLM adaptation produced consistent positive gains for both models. AfriBERTa-MLM improved Macro-F1 by +0.007 over its baseline (0.107 to 0.114) and accuracy from 0.200 to 0.207, becoming the best neural configuration overall. SwahBERT-MLM showed a larger relative improvement, gaining +0.012 Macro-F1 (0.073 to 0.085) and lifting accuracy from 0.149 to 0.171. The larger gain for SwahBERT is consistent with the expectation that a model whose news-domain pre-training is most stylistically distant from proverbs benefits most from adaptation toward the proverbial register. Despite these gains, neither adapted model approached the static baselines.

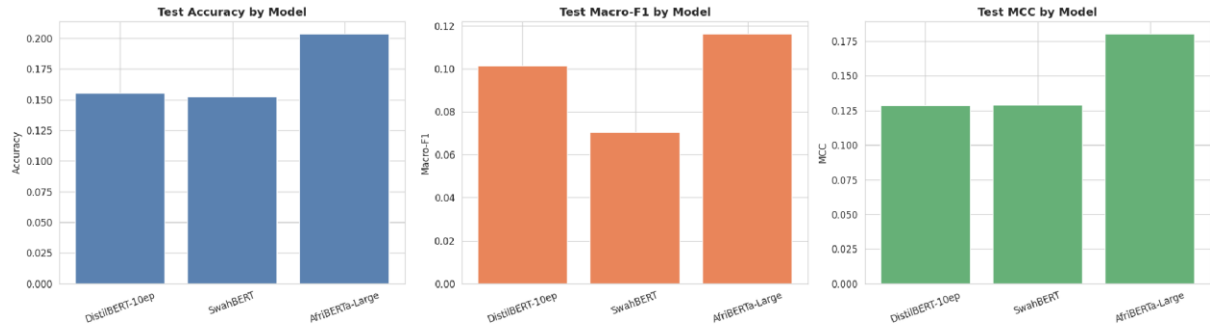


Figure 2. Test accuracy and Macro-F1 across transformer fine-tuning configurations.

Condition	Test Accuracy	Test Macro-F1	Test MCC	F1 Gain vs Baseline
AfriBERTa-Baseline	0.200	0.107	0.176	n/a
AfriBERTa-MLM	0.207	0.114	0.183	+0.007
SwahBERT-Baseline	0.149	0.073	0.123	n/a
SwahBERT-MLM	0.171	0.085	0.148	+0.012

Table 7. Task 3 results, MLM domain adaptation effect.



Figure 3. Baseline versus MLM-adapted scores for AfriBERTa and SwahBERT.

Task 4: Llama-3 Prompting

Results for Llama-3-8B-Instruct evaluated on 20 proverbs are shown in **Table 8** and **Figure 4**. Both prompting conditions produced 20 valid predictions out of 20, indicating reliable instruction-following at the output-format level. Zero-shot prompting achieved test accuracy 0.200, Macro-F1 0.211, and MCC 0.212, the highest MCC of any method in the entire study. Few-shot prompting with 10 illustrated categories performed worse on every metric (accuracy 0.150, Macro-F1 0.089, MCC 0.115). The decline under few-shot prompting is discussed in Section 5.5. Because of the small evaluation set, these figures are treated as exploratory.

Method	Valid Preds (of 20)	Test Accuracy	Test Macro-F1	Test MCC
Zero-shot	20	0.200	0.211	0.212
Few-shot (10 examples)	20	0.150	0.089	0.115

Table 8. Task 4 results, Llama-3-8B zero-shot and few-shot prompting.

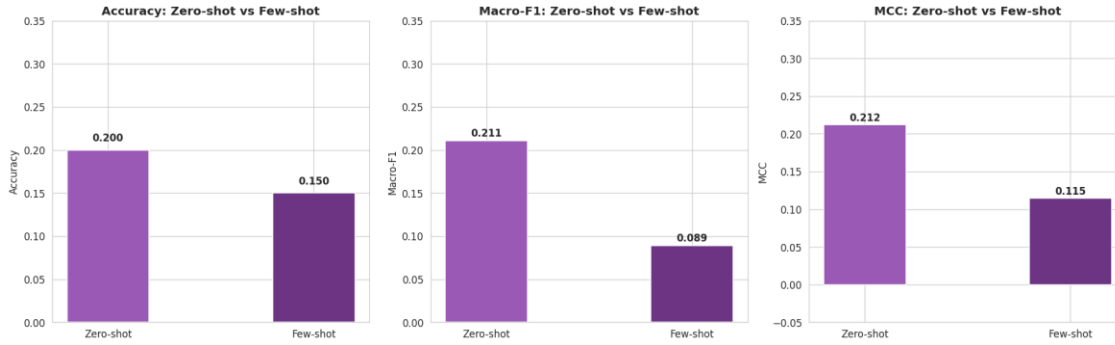


Figure 4. Llama-3 zero-shot versus few-shot prompting.

Cross-Task Summary

Table 9 consolidates the best-performing method from each task. Character n-gram TF-IDF (Task 1) achieved the highest Macro-F1 (0.225) and accuracy (0.231) of any method evaluated. The best neural result, AfriBERTa-MLM (Task 3), reached Macro-F1 0.114, roughly half the static baseline. Llama-3 zero-shot (Task 4) achieved the highest MCC (0.212) but a Macro-F1 (0.211) just below the character n-gram method, on a much smaller evaluation sample. Taken together, the results show that on this small, metaphorically dense corpus, sparse statistical and sub-word representations outperform contextual neural models, and that the metric chosen materially affects which method appears strongest.

Task	Best Method	Test Accuracy	Test Macro-F1	Test MCC
Task 1: Static Embeddings	Char N-gram TF-IDF	0.231	0.225	0.211
Task 2: Transformer Fine-tuning	AfriBERTa-Large	0.204	0.116	0.180
Task 3: MLM + Fine-tuning	AfriBERTa-MLM	0.207	0.114	0.183
Task 4: LLM Prompting	Llama-3 Zero-shot	0.200	0.211	0.212

Table 9. Best-performing method per task, cross-paradigm comparison. Best Macro-F1 in bold.

5. Discussion

With 56 categories and support ranging from 5 to over 200 samples, standard accuracy can be deceptive. The results illustrate this concretely: Llama-3 zero-shot and character n-gram TF-IDF reach almost identical accuracy (0.200 versus 0.231) but tell different stories under MCC, where Llama-3 leads (0.212 versus 0.211). Macro-F1 and MCC, which weight all categories or all confusion-matrix quadrants equally, are therefore the more trustworthy primary metrics here (Chicco and Jurman, 2020). The gap between accuracy and Macro-F1 for the transformer models, for example AfriBERTa-Large at 0.204 accuracy but only 0.116 Macro-F1, shows that their accuracy is inflated by correct predictions on high-support categories while coverage of the long tail remains poor. Per-class classification reports are provided in **Appendix B**.

The fundamental challenge of this task is the semantic gap between proverb text and category label. A proverb about a setting sun does not contain the word 'debt'; a proverb about a departing visitor does not contain the word 'friendship'. Models must learn to associate surface lexical and morphological patterns with abstract thematic categories purely from labelled examples, a form of metaphor comprehension that remains an open problem in NLP. The low absolute performance of all methods (best Macro-F1 0.225 on a 56-class problem, against a random baseline near 0.018 accuracy) reflects this difficulty.

The superiority of sparse representations over transformer models is consistent with this framing. With as few as 3 to 4 training examples per class in some categories, fine-tuning a transformer does not provide sufficient signal to reorganise its attention patterns toward proverbial semantics. Character n-gram and word-level TF-IDF, by contrast, capture category-specific lexical and sub-word regularities statistically without requiring new representational structure to be learned. That the character n-gram method outperformed every other approach confirms that morphological surface patterns carry genuine discriminative signal for this agglutinative language, and that capturing them does not require a large pre-trained model.

The results show that African-language pre-training yields a clear advantage among the transformers. AfriBERTa-Large (Macro-F1 0.116, MCC 0.180) outperformed both SwahBERT (Macro-F1 0.070) and DistilBERT-multilingual (Macro-F1 0.101). Interestingly, the monolingual SwahBERT did not beat the broader multilingual models, which suggests that for this task the breadth of African-language exposure in AfriBERTa's pre-training, covering 11 related languages, matters more than the depth of Swahili-specific training in SwahBERT. SwahBERT's news-domain pre-training is likely too stylistically distant from proverbial language to confer the expected monolingual advantage, while AfriBERTa's RoBERTa-style training over related Bantu and other African languages provides a sub-word vocabulary that aligns better with proverb morphology.

MLM domain adaptation produced consistent, positive gains for both models, supporting the domain-adaptive pre-training hypothesis at this data scale. The effect was larger for SwahBERT (+0.012 Macro-F1) than for AfriBERTa (+0.007 Macro-F1). This pattern is theoretically coherent: SwahBERT begins furthest from the proverb domain, having been pre-trained on news and discussion text, so a single self-supervised pass over the proverb corpus shifts its masked-token predictions more substantially toward the proverbial register. AfriBERTa, already exposed to a broad mix of African-language text, has less distance to close and therefore gains less.

That said, the absolute gains are small (both under 0.02 Macro-F1) and were achieved with only a single epoch of adaptation. The consistency of the direction across both models is more informative than the magnitude: it indicates a real, if modest, benefit from domain adaptation. Extended adaptation over more epochs, precluded here by computational constraints, may yield larger improvements and is a natural direction for future work.

Llama-3-8B-Instruct under zero-shot prompting was unexpectedly competitive, achieving the highest MCC of any method (0.212) and a Macro-F1 (0.211) close to the best static baseline, all without any task-specific training. With 20 of 20 predictions valid, the model reliably produced category names in the required format. This suggests that a large instruction-tuned model carries enough latent semantic knowledge to map some Swahili proverbs to abstract thematic categories, though the very small evaluation sample means these figures must be read as indicative rather than conclusive.

The most counterintuitive result is that few-shot prompting performed worse than zero-shot on every metric. One likely explanation is label bias: by illustrating only 10 of the 56 categories, the few-shot prompt may have nudged the model toward over-predicting those 10, reducing accuracy on the remaining 46. Prompt design and label-parsing are known to have outsized effects on LLM classification performance (Brown et al., 2020), and a richer few-shot design covering more categories, or a larger evaluation set, would be needed to characterise this effect reliably. The broader limitation is that the 56 category labels are abstract, culture-embedded thematic concepts that may not align cleanly with Llama-3's pre-training distribution.

6. Conclusion

This study evaluated four NLP paradigms on the classification of 5,060 Swahili proverbs into 56 semantic categories: static embedding classification, transformer fine-tuning, MLM domain adaptation, and LLM prompting. The task is representative of a broader class of NLP problems characterised by small labelled datasets, severe class imbalance, metaphorical language, and a morphologically rich target language underrepresented in standard pre-training corpora.

The central empirical finding is that sparse statistical representations outperformed all neural approaches. Character n-gram TF-IDF achieved the best overall results (test accuracy 0.231, Macro-F1 0.225, MCC 0.211), with word-level TF-IDF and pre-trained FastText close behind. The strongest neural model, AfriBERTa-Large with MLM adaptation, reached Macro-F1 0.114, roughly half the best static baseline. MLM domain adaptation produced consistent positive gains for both transformers (SwahBERT +0.012, AfriBERTa +0.007 Macro-F1). Llama-3-8B zero-shot prompting was surprisingly competitive, achieving the highest MCC of any method (0.212), and unexpectedly outperformed few-shot prompting. All methods substantially exceeded the random baseline but remain far from practical utility, underscoring the inherent difficulty of metaphor-grounded semantic classification from limited labelled data. This represents the biggest challenge encountered in this study and as I started with a proverb, it is only fit that I conclude with one “*Penye nia pana njia*”, meaning where there is a will there is a way and hope that future studies can find the way to improve on this.

Contributions

This work makes three primary contributions. First, it provides the first systematic comparison of static embeddings, BERT-style fine-tuning, domain-adaptive pre-training, and LLM prompting on Swahili proverb classification, establishing reproducible baseline performance figures for a previously unstudied task. Second, it demonstrates the practical importance of metric selection for imbalanced classification: accuracy, Macro-F1, and MCC rank the methods differently, and reporting only one would misrepresent the results. Third, it provides a reproducible experimental framework tracked via Weights & Biases and MLflow that can be extended to other low-resource African language classification tasks.

Limitations

Class imbalance and low-support categories with fewer than 10 training examples constrain what any current method can reliably learn, and several categories are effectively unlearnable at this data scale. MLM domain adaptation was run for only 1 epoch due to computational constraints, and extended adaptation may produce larger gains. Task 4's evaluation on 20 samples is insufficient for reliable metric estimation and should be treated as exploratory. Finally, the corpus was collected from a single source (Scheven, 2010), which may not fully represent the diversity of Swahili proverbial expression across regional dialects and registers.

Future Work

Several directions offer promising extensions. Data augmentation, including back-translation through a high-resource language, morphological variation, or paraphrase generation, could expand the effective training set for low-support categories. Extended MLM adaptation over multiple epochs would test whether the modest gains observed here continue to grow. A larger and more balanced few-shot prompt design, evaluated on a larger sample, would clarify the surprising zero-shot versus few-shot result for Llama-3. Parameter-efficient fine-tuning methods such as LoRA (Hu et al., 2022) would make it feasible to adapt larger LLMs within standard GPU memory constraints. Finally, extending the study to proverb datasets in other low-resource languages would test the generalisability of these findings beyond Swahili.

GitHub Repository:

<https://github.com/belinatom/NALAPROJECT>.

WeightsandBiases:

<https://wandb.ai/belinda-mziray-hochschule-luzern/nalapro-swahili/reports/NALAPRO-MODEL-EVALUATION-METRICS--VmlldzoxNzE1OTUxMA>

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Appendix

A. Label Distribution

Figure A1 shows the class distribution across all 56 semantic categories, sorted by frequency.

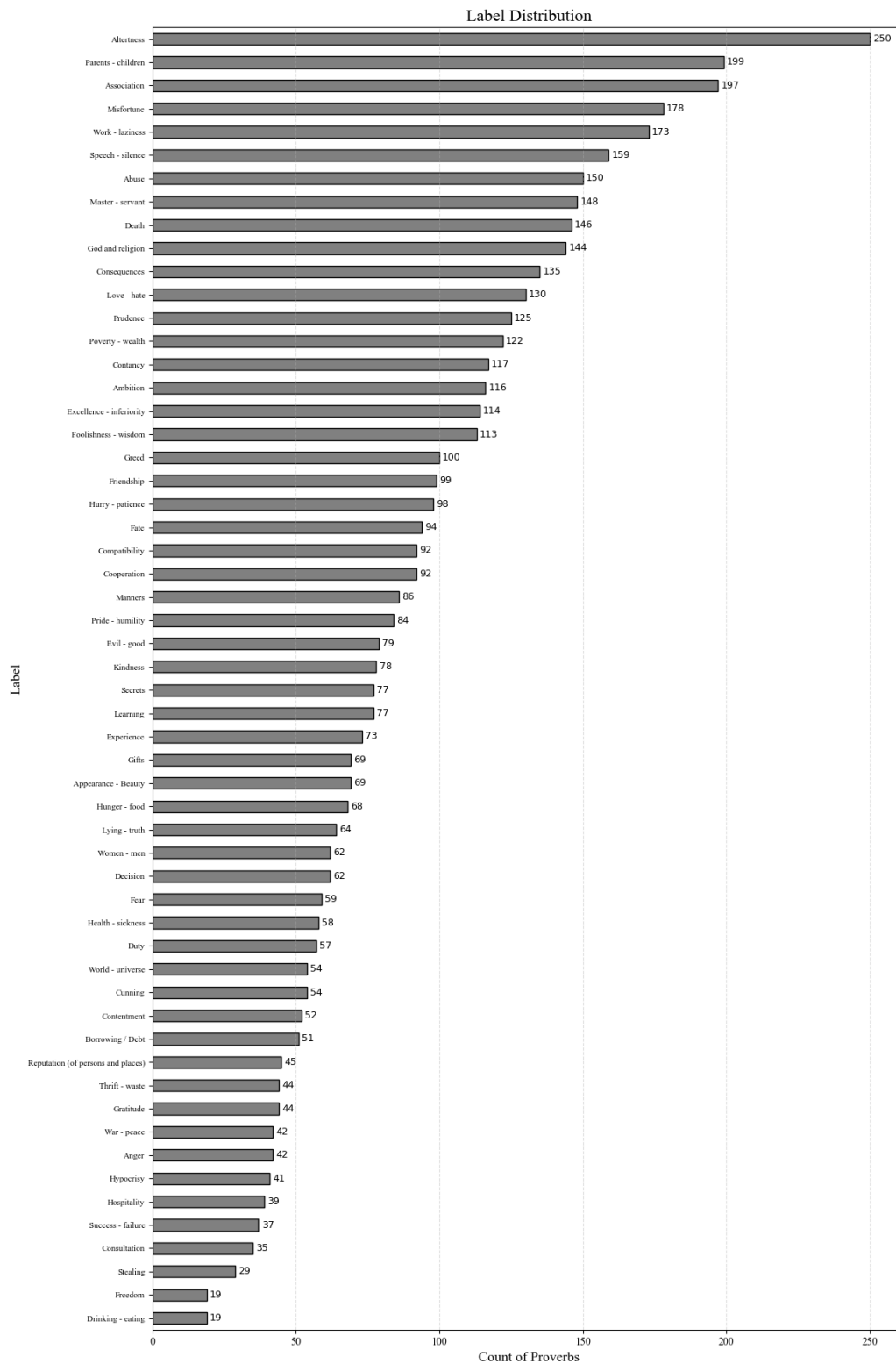
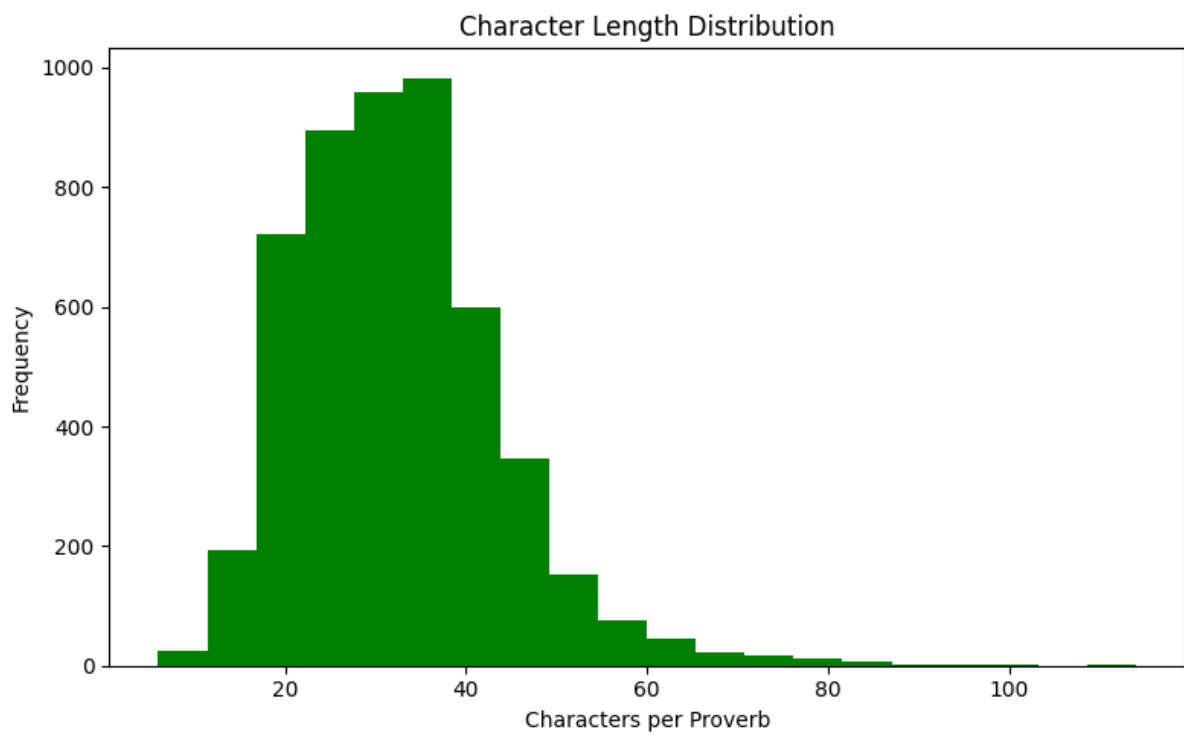
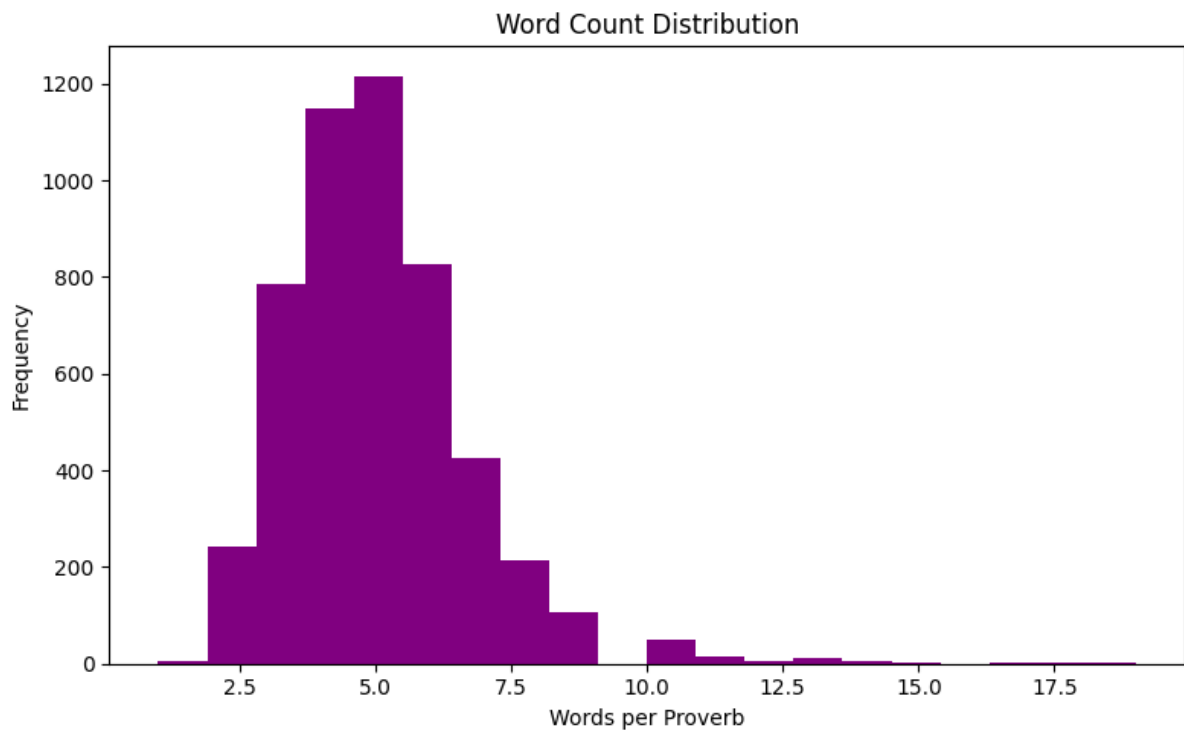
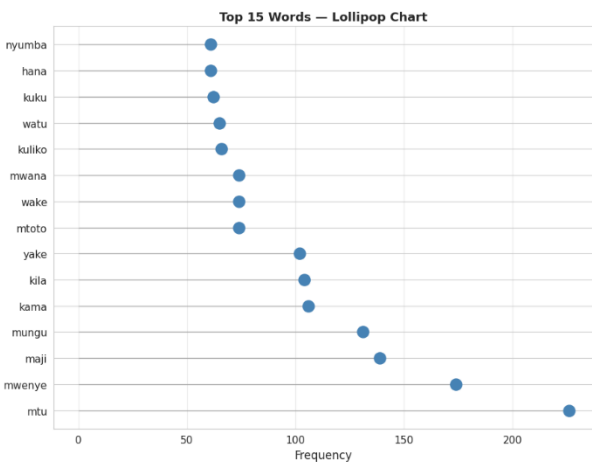
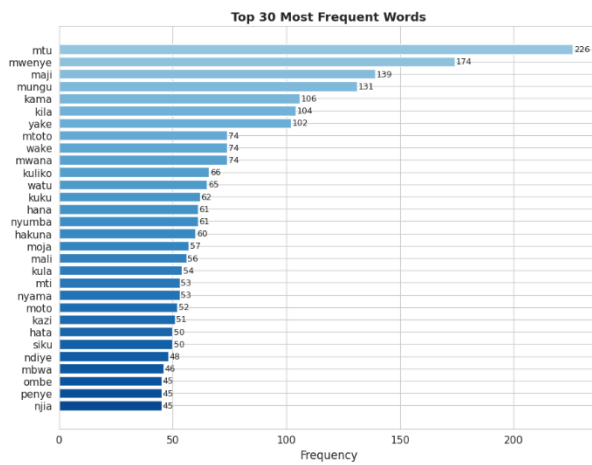
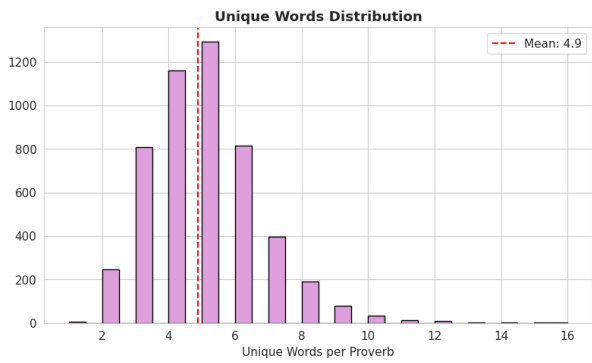
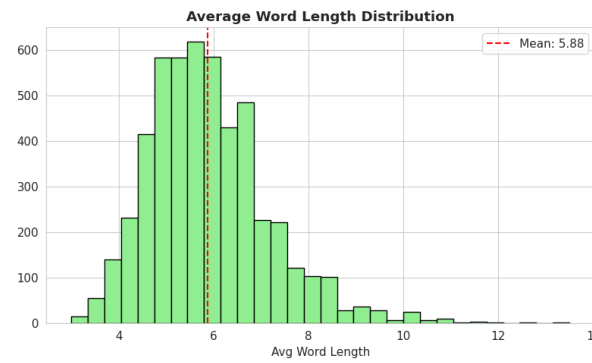
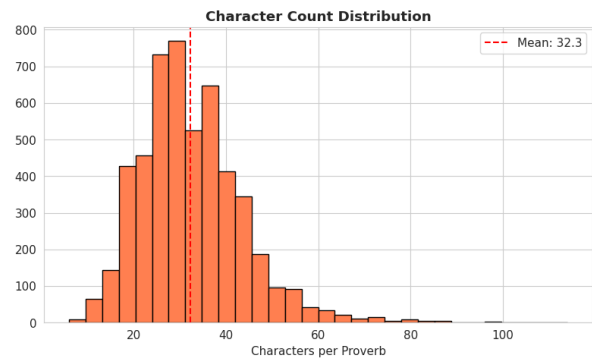
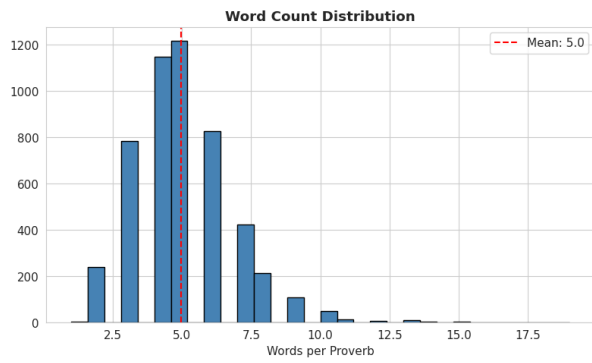


Figure A1. Sample counts per semantic category, sorted by frequency.

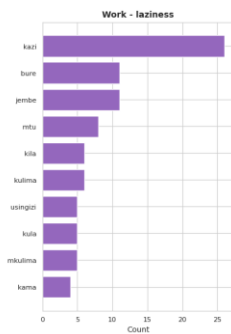
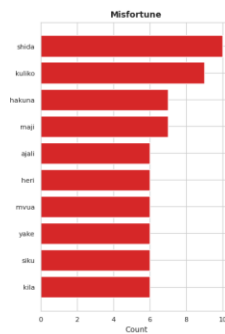
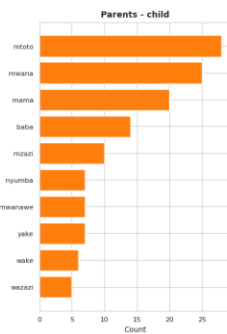
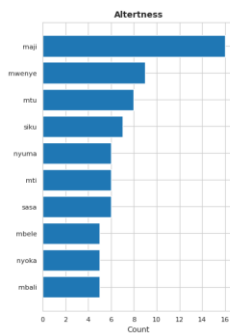
B. Dataset Summary Statistics.



Text Length Analysis



Top 10 Words per Category (Top 5 Categories)



Note on Use of AI Tools

Large language model assistance (Claude, Anthropic) was used during the preparation of this work for two purposes: editorial refinement of written content and code development support across pipeline components. All research design, methodological decisions, empirical analysis, and interpretations remain solely mine.